Shiwin Fernando Igwin Fernando

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EDUCATION

University of Texas at San Antonio | San Antonio, TX **August 2027**

B.S. in Computer Science | GPA: 3.88/4.0

- **Relevant Courses:** Principles of Cybersecurity, Database Systems, Data Science, Artificial Intelligence, Cloud Computing, Cryptography, Machine Learning, Data Mining, Secure Software Development and Analysis

TECHNICAL SKILLS

Languages (by proficiency): Python, SQL, Java, C, HTML, CSS, JavaScript, Go, Kotlin

Frameworks: Node.js, React.js, PyTorch, TensorFlow, OpenCV, Matplotlib, NumPy, Pandas, scikit-learn, Flask, FastAPI, Streamlit, LangChain, LangGraph, CrewAI, SQLAlchemy, Plotly

Tools: AWS EC2, Oracle Cloud, Google Cloud, PostgreSQL, Git/Github, MongoDB, Tableau, PowerBI, ServiceNow, Android Studio, MySQL Workbench, Cursor, Visual Studio Code, Jupyter Notebook, Jupyter Lab, n8n, Docker, ChromaDB

RELEVANT EXPERIENCE

University of Texas at San Antonio's RISE Lab | San Antonio, TX **June 2026 - Present**

Undergraduate Research Assistant

- Testing neural network and transformer models for project focused in AI-generated personalized music interventions aimed at reducing mental distress in subjects
- Assisting in experiments in AI model weight regeneration using mathematical equations and evolutionary methods

UTS (University Tech Solutions) Bold Careers | San Antonio, TX **May 2025 - August 2025**

ServiceNow Data Analyst Intern

- Constructed dashboards and reports from examining 750+ rows of ITSM ticket table data on daily average using ServiceNow platform for clients to help keep track of systems workflows
- Customized existing AI agents in ServiceNow Studio to detect data anomalies and automate help-desk configuration workflows
- Mapped and planned organizational objectives and business outcomes to measurements (KPI trees) before technical configuration

San Antonio Museum of Science and Technology | San Antonio, TX **Feb 2023 - May 2023**

Research Intern and Liaison

- Conducted surveys with museum guests and co-workers and collected at least 50 responses for data analysis research project on impact of computerization and workflow automation in blue-collar demographic areas of San Antonio
- Engineered an ensemble ML pipeline (Random Forest + Gradient Boosting on TF-IDF survey features) to classify sentiment, then used Prophet to forecast 2024–2025 trend direction in East Side San Antonio's attitudes toward technological advancement

PROJECTS

Nocturne | Python, FastAPI, LangGraph, PostgreSQL, pgvector, Gemini, Groq, Langfuse, GitHub Actions **November 2026**

- Built an LLM memory agent with a LangGraph sleep-time consolidation pipeline (3 of 7 planned sub-agents complete) that extracts facts, deduplicates, and resolves contradictions across conversation history
- Designed a bitemporal PostgreSQL/pgvector schema across 4 tables with full version history and 0 hard deletes, making 100% of memory edits auditable and reversible.
- Developed hybrid vector and keyword retrieval with Langfuse tracing and CI on 2 platforms, running at $0 cost on free tiers; now building a LongMemEval benchmark against a naive RAG baseline

Axolotl | Python, LangGraph, Kafka, FastAPI, Pandera, DuckDB, PostgreSQL, Docker, GitHub Actions, React, TypeScript **October 2026**

- Built a multi-agent LangGraph system that detects, diagnoses, and repairs live data pipeline failures, then opens verified GitHub pull requests for human review
- Detected schema drift in a median of 1.3 seconds and quarantined bad records in a Kafka dead-letter queue, recovering 100% of 6,854 records with zero data loss
- Validated every large language model (LLM) patch in isolated Docker sandboxes against unit tests, held-out data, and live traffic, with 0 false fixes across 45 benchmark runs
- Ran a chaos engineering benchmark of 9 failure scenarios at $0.001 per incident, correctly escalating 100% of outages and ignoring 100% of harmless schema changes